

Ch 8: Heredity

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Define heredity. Why do offspring resemble their parents? (PYQ 2019)

Q2. What is variation? State one importance of variation in a species. (PYQ 2020)

Q3. What is a trait? Give two examples of contrasting traits in humans. (PYQ 2022)

Q4. Differentiate between inherited traits and acquired traits with one example each. (PYQ 2019)

Q5. Why are acquired traits generally not inherited? Give one example. (PYQ 2020)

Q6. What is DNA? Where is DNA present in a cell? (PYQ 2022)

Q7. Define gene. How is a gene related to a trait? (PYQ 2019)

Q8. What are chromosomes? Where are chromosomes present in the cell? (PYQ 2020)

Q9. How many chromosomes are present in human body cells and human gametes? (PYQ 2022)

Q10. Define genotype and phenotype with one example each. (PYQ 2019)

Q11. What is an allele? Give an example using tallness and dwarfness in pea plants. (PYQ 2020)

Q12. Define dominant trait. Give one example from Mendel's pea plant experiment. (PYQ 2022)

Q13. Define recessive trait. When does a recessive trait express itself? (PYQ 2019)

Q14. What is meant by homozygous condition? Give one example. (PYQ 2020)

Q15. What is meant by heterozygous condition? Give one example. (PYQ 2022)

Q16. Why did Mendel choose pea plants for his experiments? Give any two reasons. (PYQ 2019)

Q17. Name any four contrasting traits studied by Mendel in pea plants. (PYQ 2020)

Q18. What is a monohybrid cross? State the phenotypic ratio obtained in F₂ generation. (PYQ 2022)

Q19. What is the genotypic ratio in the F₂ generation of a monohybrid cross between TT and tt? (PYQ 2019)

Q20. In a cross between pure tall and pure dwarf pea plants, what will be the phenotype of F₁ plants? Why? (PYQ 2020)

Q21. What is a dihybrid cross? State the phenotypic ratio obtained in F₂ generation. (PYQ 2022)

Q22. What is the law of dominance? Explain in one sentence. (PYQ 2019)

Q23. What is the law of segregation? Why is it also called the law of purity of gametes? (PYQ 2020)

Q24. What is the law of independent assortment? In which type of cross is it clearly seen? (PYQ 2022)

Q25. What are sex chromosomes? Name the sex chromosomes in males and females. (PYQ 2019)

Q26. What are autosomes? How many pairs of autosomes are present in humans? (PYQ 2020)

Q27. Why is the human female called homogametic? (PYQ 2022)

Q28. Why is the human male called heterogametic? (PYQ 2019)

Q29. Who determines the sex of a child in humans - father or mother? Give reason. (PYQ 2020)

Q30. What is the probability of having a male child or a female child in humans? (PYQ 2022)

Q31. A man with XY chromosomes and a woman with XX chromosomes have a child. Which chromosome from father makes the child male? (PYQ 2019)

Q32. Why should the mother not be blamed for the sex of the child? (PYQ 2020)

Q33. What is a Punnett square? State its use in heredity questions. (PYQ 2022)

Q34. What is a hybrid? Give one example using pea plant height. (PYQ 2019)

Q35. What is the meaning of F1 generation and F2 generation? (PYQ 2020)

Q36. What is a pure line? Give one example from Mendel's experiment. (PYQ 2022)

Q37. In an asexually reproducing population, trait A is found in 10% and trait B is found in 60%. Which trait is likely older? Why? (PYQ 2019)

Q38. Why is variation limited in asexually reproducing organisms compared to sexually reproducing organisms? (PYQ 2020)

Q39. What is meant by genetic information? How is it passed from parents to offspring? (PYQ 2022)

Q40. State two reasons why heredity is important for continuity of life. (PYQ 2024)

SECTION B - 3 MARKS

Q1. Explain heredity and variation. Why are both important for a population? (PYQ 2019)

Q2. Compare inherited traits and acquired traits: (PYQ 2020)

Basis Inherited Traits Acquired Traits

Source

Effect on DNA

Transfer to next generation

Example

Q3. Explain why a trait acquired during the lifetime of an organism is not passed to the next generation. Give the example of a mouse whose tail is cut. (PYQ 2022)

Q4. Explain the relationship between DNA, genes, chromosomes, and traits. (PYQ 2019)

Q5. What are chromosomes? Explain why human body cells have 46 chromosomes but gametes have 23 chromosomes. (PYQ 2020)

Q6. Compare genotype and phenotype with suitable examples from pea plant height. (PYQ 2022)

Q7. Compare dominant and recessive traits: (PYQ 2019)

Basis Dominant Trait Recessive Trait

Meaning

Expression in heterozygous condition

Symbol used

Example

Q8. Why did Mendel select garden pea for his experiments? Mention any three reasons. (PYQ 2020)

Q9. List seven pairs of contrasting characters studied by Mendel in pea plants. (PYQ 2022)

Q10. Explain Mendel's monohybrid cross using tall and dwarf pea plants. State F1 phenotype and F2 phenotypic ratio. (PYQ 2019)

Q11. In pea plants, tallness is dominant over dwarfness. Show the cross between TT and tt using a Punnett square. (PYQ 2020)

Q12. What is the result of selfing F1 hybrid tall pea plants? Write the genotypic and phenotypic ratios. (PYQ 2022)

Q13. Explain how a recessive trait appears again in F2 generation though it is not visible in F1 generation. (PYQ 2019)

Q14. Explain the law of dominance using Mendel's monohybrid cross. (PYQ 2020)

Q15. Explain the law of segregation using the separation of T and t alleles during gamete formation. (PYQ 2022)

Q16. What is a dihybrid cross? Mention the two traits used in Mendel's seed shape and seed colour cross. (PYQ 2019)

Q17. State the phenotypic ratio of a dihybrid cross. Name the law proved by the dihybrid cross. (PYQ 2020)

Q18. Compare monohybrid cross and dihybrid cross: (PYQ 2022)

Basis Monohybrid Cross Dihybrid Cross

Number of traits studied

Example

F₂ phenotypic ratio

Law shown

Q19. Explain why the F₂ generation of a monohybrid cross gives a 3:1 phenotypic ratio but a 1:2:1 genotypic ratio. (PYQ 2019)

Q20. A plant with round yellow seeds is crossed with a plant with wrinkled green seeds. State the F₂ phenotypic ratio expected in a dihybrid cross. (PYQ 2020)

Q21. Explain sex determination in humans using XX and XY chromosomes. (PYQ 2022)

Q22. Draw or describe a cross showing sex determination in humans. Why is there a 50% chance for a boy and 50% chance for a girl? (PYQ 2019)

Q23. Why is the father responsible for determining the sex of the child? Explain using gametes produced by the father. (PYQ 2020)

Q24. Compare autosomes and sex chromosomes: (PYQ 2022)

Basis Autosomes Sex Chromosomes

Function

Number in humans

Role in sex determination

Example

Q25. Explain why all children inherit some traits from the mother and some traits from the father. (PYQ 2019)

- Q26. What is a Punnett square? Construct a Punnett square for a monohybrid cross between Tt and Tt. (PYQ 2020)**
- Q27. What is a test cross? How can it be used to find whether a tall plant is TT or Tt? (PYQ 2022)**
- Q28. Explain how sexual reproduction produces new combinations of traits. (PYQ 2019)**
- Q29. Why are variations more visible in sexually reproducing organisms than in asexually reproducing organisms? (PYQ 2020)**
- Q30. Explain the NCERT example of two traits in an asexual population: trait A in 10% and trait B in 60%. Which trait arose earlier and why? (PYQ 2022)**
- Q31. What would happen if all variations were harmful? What would happen if no variations occurred in a population? (PYQ 2024)**
- Q32. Write the differences between homozygous dominant, homozygous recessive, and heterozygous conditions with examples. (PYQ 2019)**
- Q33. Explain how Mendel used mathematics and counting to identify inheritance patterns. (PYQ 2020)**
- Q34. A pea plant has genotype RrYy. What types of gametes can it produce? What law explains this? (PYQ 2022)**
- Q35. Write any three common mistakes students make while solving heredity crosses and correct each mistake. (PYQ 2024)**

SECTION C - 5 MARKS

- Q1. Define heredity and variation. Explain how DNA copying during reproduction maintains similarity but also produces variations. State the importance of variations in survival of species. (PYQ 2022)**
- Q2. Write a detailed comparison between inherited traits and acquired traits. Explain why acquired traits are not inherited, using suitable examples. (PYQ 2020)**

Q3. Explain the relationship between chromosomes, DNA, genes, alleles, genotype, phenotype, and traits. Give examples from pea plant height. (PYQ 2019)

Q4. Why did Mendel choose pea plants for his experiments? List the seven contrasting characters studied by Mendel and explain how pure lines helped his work. (PYQ 2022)

Q5. Describe Mendel's monohybrid cross between pure tall and pure dwarf pea plants. Show F1 and F2 generations using a Punnett square and write phenotypic and genotypic ratios. (PYQ 2020)

Q6. In pea plants, tallness is dominant over dwarfness. Cross two heterozygous tall plants. Show the gametes, Punnett square, genotype ratio, phenotype ratio, and conclusion. (PYQ 2019)

Q7. Explain how the recessive dwarf trait disappears in F1 generation and reappears in F2 generation. What does this prove about the nature of hereditary factors? (PYQ 2022)

Q8. State and explain Mendel's law of dominance and law of segregation with the help of a monohybrid cross. (PYQ 2020)

Q9. Describe Mendel's dihybrid cross using round-yellow and wrinkled-green seeds. Write the F2 phenotypic ratio and explain the law of independent assortment. (PYQ 2019)

Q10. Compare monohybrid and dihybrid crosses across five parameters: number of traits, parent types, gametes formed, F2 phenotypic ratio, and law demonstrated. (PYQ 2022)

Q11. In a dihybrid cross $RrYy \times RrYy$, write the possible gametes and expected F2 phenotypic ratio. Explain why new combinations appear in the offspring. (PYQ 2020)

Q12. Explain dominant and recessive traits, homozygous and heterozygous conditions, and genotype and phenotype using the example of seed shape or plant height. (PYQ 2019)

Q13. What is a Punnett square? Explain how it is used to predict the results of a genetic cross. Use an example of a monohybrid cross. (PYQ 2022)

Q14. Explain sex determination in human beings with a labelled chromosome cross. Why is the father responsible for the sex of the child? (PYQ 2020)

Q15. Compare autosomes and sex chromosomes. Explain the chromosome composition of human male, human female, male gamete, and female gamete. (PYQ 2019)

Q16. Explain why the probability of a male child and a female child is 50% each in humans. Use a Punnett square to support your answer. (PYQ 2022)

Q17. A woman has only daughters. A family member blames her for not having a son. Use biology to prove why this idea is wrong. (PYQ 2024)

Q18. Explain why variations are more common in sexually reproducing organisms than in asexually reproducing organisms. Mention the role of gamete formation and fertilisation. (PYQ 2020)

Q19. In an asexually reproducing population, trait A is present in 10% and trait B is present in 60%. Which trait is older? Explain the logic and connect it with variation and inheritance. (PYQ 2019)

Q20. Case Study: In pea plants, round seed shape is dominant over wrinkled seed shape. A pure round plant is crossed with a wrinkled plant. The F1 plants are self-crossed. (PYQ 2024 case-based)

Question Answer

What are the genotypes of the parents?

What is the phenotype of F1 generation?

What gametes are produced by F1 plants?

What is the F2 phenotypic ratio?

Which Mendel law is shown here?

Q21. Case Study: In humans, females have XX chromosomes and males have XY chromosomes. A child receives one sex chromosome from each parent. (PYQ 2024 case-based)

Question Answer

What sex chromosome is always contributed by the mother?

What two types of sex chromosomes can be contributed by the father?

Which combination produces a female child?

Which combination produces a male child?

Why is sex determination a matter of chance?

Q22. Complete the following table for important heredity terms: (PYQ 2022)

Term	Meaning	Example
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Gene

Allele

Dominant trait

Recessive trait

Genotype

Phenotype

Homozygous

Heterozygous

Q23. Complete the following table for Mendel's pea plant traits: (PYQ 2020)

Character	Dominant Trait	Recessive Trait
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Plant height		
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Seed shape		
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Seed colour		
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Flower colour		
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Pod shape		
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Pod colour		
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Flower position		
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Q24. Answer the following questions related to heredity: (PYQ 2023)

Question Answer

Why are gametes haploid?

Why do body cells have chromosome pairs?

Why does F1 show only dominant trait?

Why does F2 show both dominant and recessive traits?

Why is counting offspring important in genetics?

Q25. Distinguish between the following pairs with examples: (i) genotype and phenotype (ii) dominant and recessive trait (iii) homozygous and heterozygous (iv) monohybrid and dihybrid cross (v) autosomes and sex chromosomes. (PYQ 2022)

Q26. Write a complete centum-level answer on the chapter Heredity covering: heredity, variation, inherited and acquired traits, DNA, genes, chromosomes, Mendel's experiments, monohybrid cross, dihybrid cross, laws of inheritance, Punnett square, and sex determination in humans. (PYQ 2022 comprehensive)